

**Project Development Phase
Model Performance Test**

Date	7 February 2026
Team ID	LTVIP2026TMIDS75625
Project Name	Weather-Based Prediction of Wind Turbine Energy Output: A Next-Generation Approach to Renewable Energy Management
Maximum Marks	10 Marks

Model Performance Testing – Wind Turbine Energy Prediction:

S.No	Parameter	Values	Screenshot
1	Metrics	Regression Model: MAE – 120.5 MSE – 18,200.3 RMSE – 134.9 R² Score – 0.95 Classification Model (if applied): Confusion Matrix – N/A (not applicable, regression problem) Accuracy Score – N/A Classification Report – N/A	Scatter plot of Actual vs Predicted Power
2	Tune the Model	Hyperparameter Tuning – Adjusted n_estimators (100 → 200), max_depth tested for optimization Validation Method – Train/Test Split (80/20) with R² evaluation	Grid search or scatter plot